
University Ethics Form Version 3

Applicant Details (922)

Is this approval for a: (11240)

Type: (!/list-dropdown)

A2 - Student Project

What type of degree programme is being studied? (11319)

Type: (!/list-dropdown)

A2 - Postgraduate Taught (e.g. MSc / MA)

Name of Principal Researcher (11241)

Type: (S/text-short)

Hoang Nguyen Lai

Please enter your email address (11258)

Type: (S/text-short)

c5053984@newcatle.ac.uk

Please select your school/academic unit (11242)

Type: (!/list-dropdown)

A1 - School of Computing

Please enter the module code (11243)

Type: (S/text-short)

CSC8639

Please enter your supervisor's email (11259)

Type: (S/text-short)

Jichun.Li@newcastle.ac.uk

Please select your supervisor's school/unit: (11244)

Type: (!/list-dropdown)

A1 - School of Computing

Project Details (923)

Project Title (11245)

Type: (S/text-short)

An AI-Enabled Digital Twin for EV Charging: Behavioural Simulation, Policy Evaluation, and Pricing Optimisation

Project Synopsis (11257)

Type: (T/text-long)

An AI-powered digital twin of an EV charging network will be built using historical session data, including timestamps, energy consumption, duration, charger type, utilization, location, and carbon intensity. The system uses agent-based simulation, modelling EV drivers as heterogeneous agents making decisions about when, where, and how much to charge, while charging stations are treated as capacity-constrained resources. This framework enables "what-if" policy evaluation under realistic behavioral dynamics and uncertainty across three scenarios: EV adoption growth (rising arrival rates and demand), time-of-use pricing (hour- or day-varying price signals), and carbon-aware incentives that nudge drivers toward lower-carbon charging periods. A final optimisation stage, employing reinforcement learning or search-based methods, automatically learns pricing and incentive strategies designed to balance competing objectives, namely smoothing grid load, maximising station utilisation, minimising user waiting times, and reducing CO₂ emissions.

Project start date (11260)

Type: (D/date)

01/04/2026

Project end date (11261)

Type: (D/date)

29/07/2026

Is the project externally funded? (11262)

Type: (!/list-dropdown)

A3 - No

Does your project involve collaborators outside of the University? (11265)

Type: (Y/yes-no)

No

Existing Ethics, Sponsorship & Responsibility (930)

Has ethical approval to cover this proposal already been obtained? (11267)

Type: (Y/yes-no)

No

Will anyone be acting as sponsor under the NHS Research Governance

Framework for Health and Social Care? (11270)

Type: (Y/yes-no)

No

Do you have a Newcastle upon Tyne Hospitals (NUTH) reference? (11272)

Type: (Y/yes-no)

No

Will someone other than you (the principal investigator) or your supervisor (for student projects) be responsible for the conduct, management and design of the research? (11274)

Type: (Y/yes-no)

No

Animals (I) (924)

Please note that this question changed on 01/06/25 to capture activity with all animals

Does your proposed research involve the use of non-human animals i.e. the observation, capture or manipulation of animals OR their tissues (i.e. materials from animal such as skin, blood, bone, and other bodily fluids)?

(11246)

Type: (Y/yes-no)

No

NHS, Health & Social Care: Facilities, Staff & Patients (I) (925)

Will the study involve participants recruited by virtue of being NHS patients or service users, their dependents, their carers or human tissues or the use of NHS & Health/Social Care Facilities or otherwise require REC approval? (11247)

Type: (Y/yes-no)

No

Human Participants in a Non-Clinical Setting (I) (926)

Does the research involve human participants e.g. use of questionnaires, focus groups, observation, surveys or lab-based studies involving human participants?

(11249)

Type: (Y/yes-no)

No

Data (I) (927)

Does the research involve the viewing, usage or transfer of sensitive data or personal data as defined by the [General Data Protection Regulation \(GDPR\)](#) or data governed by statute such as the [Official Secrets Act 1989](#) / [Terrorism Act 2006](#) , commercial contract or by convention e.g. client confidentiality? (If you are unsure please tick YES and complete the sub-questions). (11251)

Type: (Y/yes-no)

No

Environment and Responsible Research Innovation (I) (928)

Will the study potentially cause a direct or indirect negative impact on local communities, cultural heritage or the environment? (11253)

Type: (Y/yes-no)

No

International Projects (I) (929)

Will the research be conducted outside of the UK or [European Economic Area \(EEA\)](#) , or will it involve international collaborators outside the EEA? (11255)

Type: (Y/yes-no)

No

Next Steps (931)

Based on your responses your project has been categorised as (ethically) low risk and no further review is required before you start work. You will receive a formal approval email on submission of this form. Should your project change you may need to apply for new ethical approval. (11282)

Type: (X/boilerplate)

Supporting Documentation (940)

**Please upload any documents (not uploaded elsewhere in the application)
which you think are relevant to the consideration of your application. (11308)**

Type: (/upload-files)

Summary and Submission (941)

Thank you for completing the University's Ethical Review Form. Based on your answers the University is satisfied that your project has met its ethical expectations and grants its ethical approval.

Please be aware that if you make any significant changes to your project then you should complete this form again as further review may be required. Confirmation of this decision will be emailed to you.

Please complete the declaration to submit your application.

Declaration

I certify that:

(11314)

Type: (M/multiple-opt)

the information contained within this application is accurate. (11441)

the research will be undertaken in line with all appropriate, University, legal and local standards and regulations. (11442)

I have attempted to identify the risks that may arise in conducting this research and acknowledge my obligation to (and rights of) any participants. (11443)

no work will begin until all appropriate permissions are in place. (11444)

Projects with low risks. All. Confirmation email.

(11338)

Type: (S/text-short)

Based on your answers, the University Ethics Committee grants its approval for you to start working on your project. Please be aware that if you make any significant changes to your proposal then you should complete this form again, as further review may be required. This confirmation may be used within a research portfolio as evidence of ethical approval. Please note: this confirmation will be the only correspondence you should expect to receive as evidence of ethical approval. There will be no other confirmation provided. You may now proceed with research. If you have any queries, please review the internal and external ethics FAQ pages before contacting res.policy@ncl.ac.uk.